Spatiotemporal Relighting in 4D Gaussian Splatting

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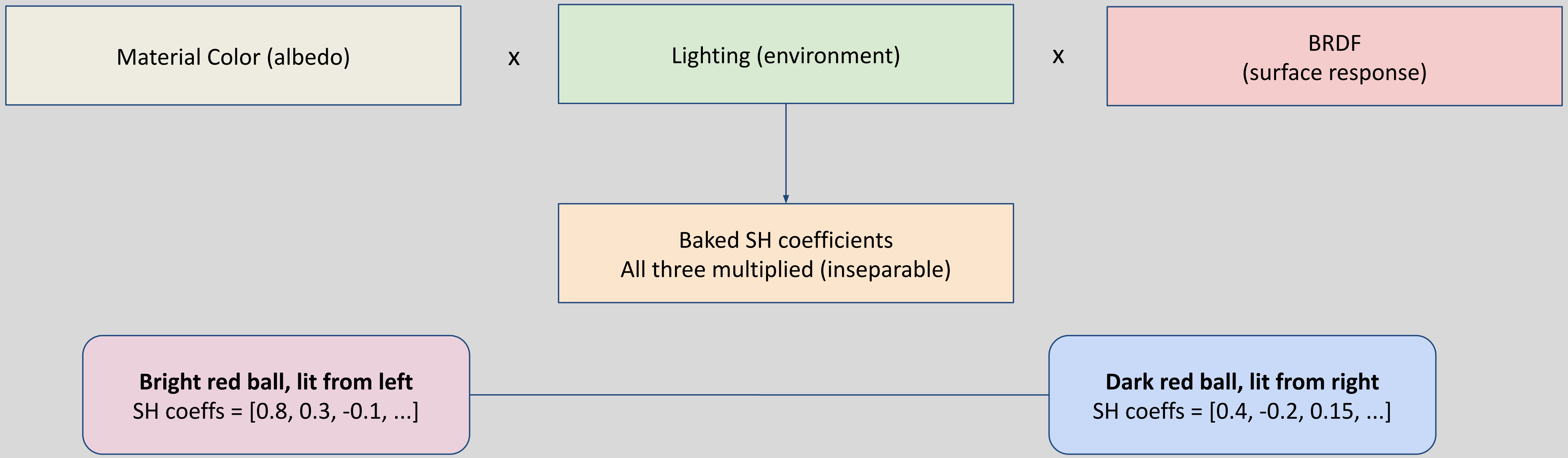
Motivation

- Compositing objects into 3D Gaussians is easy (merging point clouds), but they don't account for being rendered under different lighting conditions
  - Lighting effects are baked into spherical harmonic coefficients - not as easy as swapping out HDRs
- 1 Gaussian = 16 SH coefficients / channel



- Substantial body of research for 3D Gaussian Relighting, **but none for 4D Gaussian Relighting**
  - **Lighting undergoes temporal deformation** alongside spherical harmonic entanglement

Contributions

Shreyas:
- Writing optimization pipeline for spherical harmonic coefficient backpropagation
- Writing video rendering pipeline for 3DGS and 4DGS
- Researching assets and building custom blender scenes for simulatinig indoor/outdoor lighting differences

Ranveer:
- Writing custom masking script for scene composition
- Created various composite scenes using Hypernerf + Neu3D
- Testing Transplat on 4DGS scenes
- Attempted shadow generation for improved scene harmonization

Previous related works:

Instruct-4DGS (Kwon et al., 2025)

Instruction-guided dynamic scene editing with 4D Gau

4D-Editor (Jiang et al., 2025

Interactive object-level editing in dynamic NeRFs

D-NeRF (Pumarola et al., 2021)

Neural radiance fields for dynamic scenes

Diff4Splat (Sandstrom et al., 2025)

Controllable 4D scene generation with dynamic Gauss

GaussianCut (Jain et al., 2024)

Interactive segmentation of Gaussian splats

4D Gaussian Splatting (Wu et al., 2024)

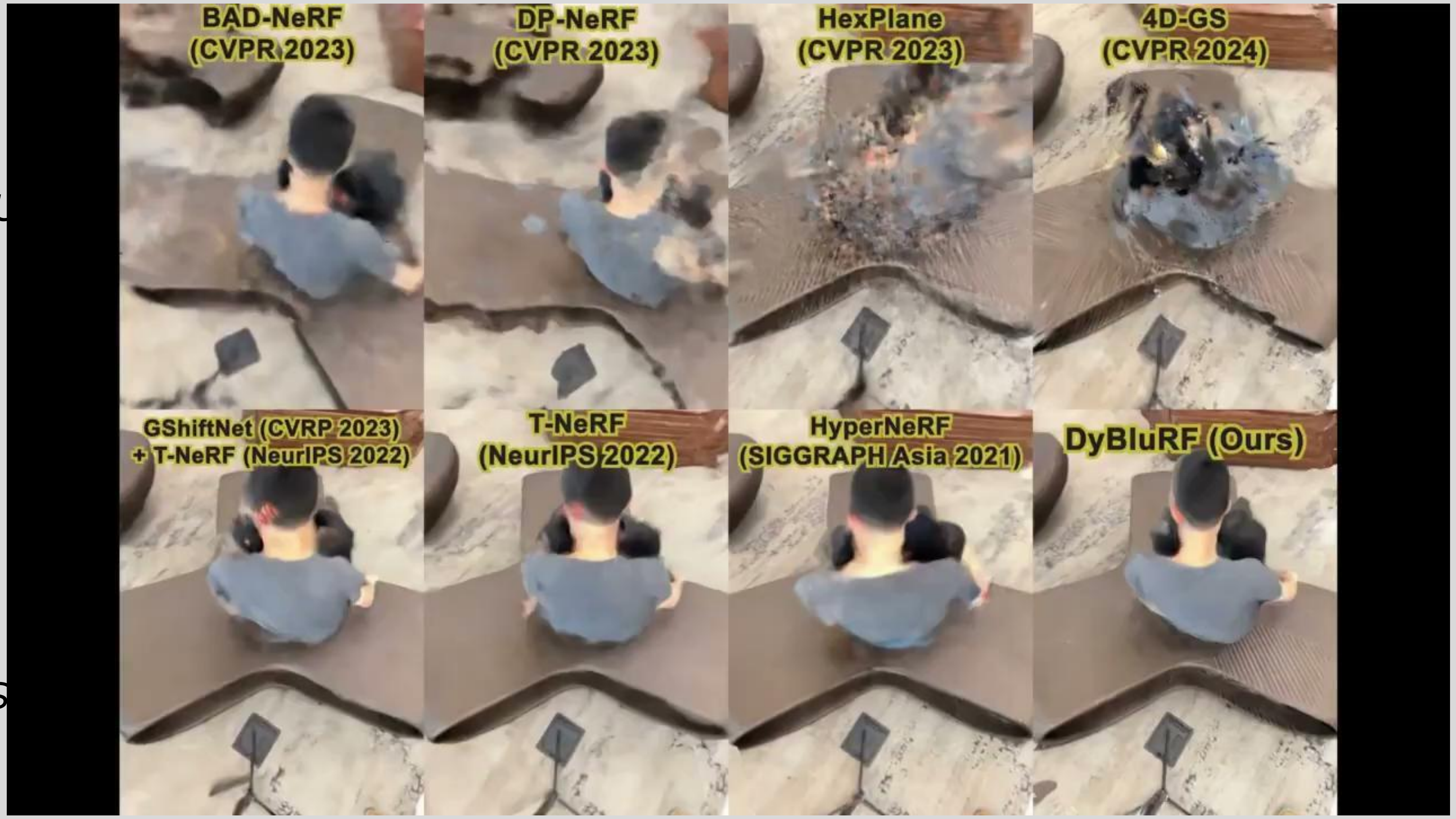
Real-time dynamic scene rendering with Gaussians

Dynamic-Editor (Lee et al., 2025)

Training-free text-driven 4D editing on 4DGS

Transplat (Yu et al., 2025)

Relighting of composite 3DGS scenes



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SA4D

- Segments objects in dynamic 4D Gaussian Splatting scenes using simple prompts (e.g., clicks or boxes)
- Uses tracking/segmentation priors from video models to supervise segmentation without manual 4D labels
- Enables object-level editing (remove, recolor, isolate) in dynamic neural scenes
- Lighting is not context-aware from scene A to scene B (cooked into segmented Gaussian field)
  - daytime scenes preserve daytime light
  - integrating daytime scene -> night is difficult

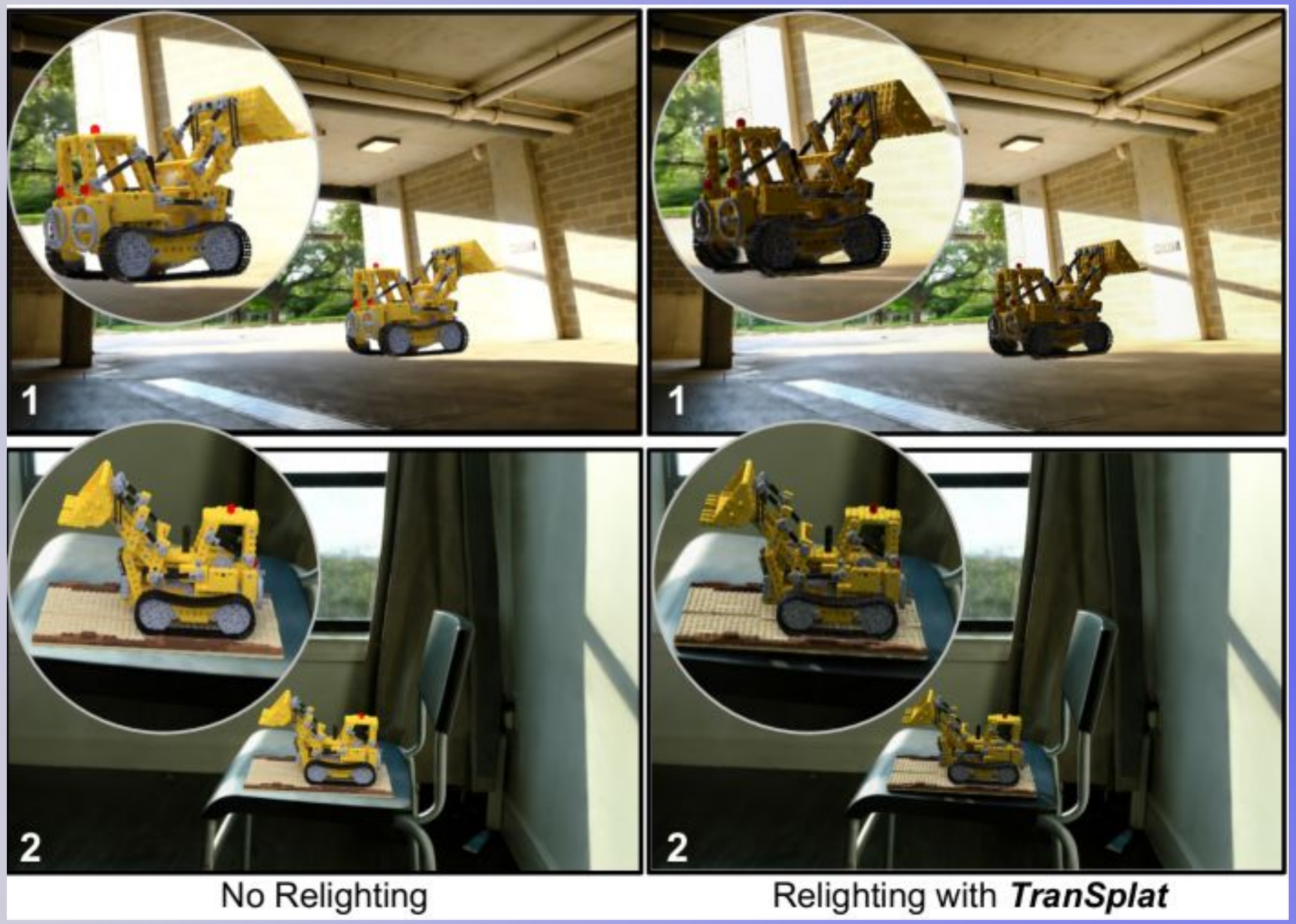
TranSplat

Closed-form SH transfer:

Compute ratio between the old and new environment lighting and scaling each Gaussian's SH coefficients by that ratio

Assumes

1. Each Gaussian is a flat surfel with a meaningful surface normal
2. SH coefficients are a **clean decomposition** of irradiance * BRDF
3. Doesn't work for extreme lighting changes



4DGS Breaks This

Interesting research direction:

On a 4D Gaussian scene that has an object from another scene transposed on top of it, how can we improve lighting for objects that come from a scene with poor lighting while preserving spatio-temporal consistency?

Roadmap

Week 1-2

1. Successfully generate 4D scenes with a low-lighting object transposed onto a bright scene
2. Improve SA4D masking for the chosen object

Weeks 3-4

1. Create an updated view using a guess based on an average of Harmonizer vectors from different views
2. Feed the 4D render and binary mask into an intrinsic harmonization model, verifying relighting

Weeks 5-6

1. Optimize for improving lighting of only the object; train end-to-end loss to pixel pipeline
2. Ensure lighting improvements are consistent across camera views and time; investigate other kinds of loss (diffusion flow; temporal) for stability
3. Compare to Blender and Transplat

Weeks 7-8

1. Stretch Goals of shadow, multi objects and reflective lighting
2. Perform quantitative analysis
3. Finalize report + generate video comparisons

Method:

Trained 4DGS (Scene A)
(SH coefficients, deformation network, Gaussian positions/scales/opacities)

Object Mask
(which of the N Gaussians are the mismatched object)

Training Camera Poses + Timesteps (2D renders)

Harmonizer Target Source
(either a model or precomputed images)

Pre-Processing

STEP 1: RENDER COMPOSITE
For each (view, timestep)
1. Render full scene -> image
2. Render object mask -> 2D mask
3. Feather mask using Gaussian blur

STEP 2: GENERATE TARGET IMAGES
For each (view, timestep)
1. Harmonize the masked area relative to the background
2. Get harmonizer target at same view/timestep

Save source, target pairs to disk (~200 needed)

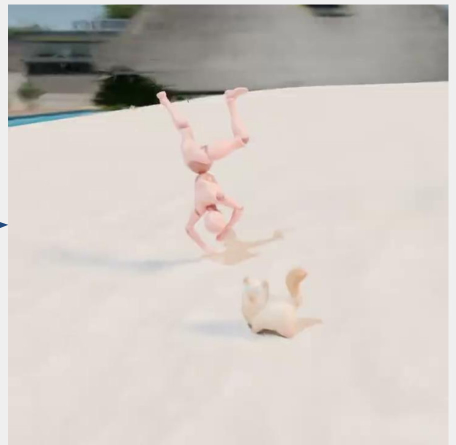
Optimize Spherical Coefficients

Initialize: ΔSH = 0 for each object Gaussian (ΔSH has same shape [N_object, 16, 3])

Each iteration:

1. **Pick a random (camera, timestep)** pair from the cache
2. **Optimization:** SH_modified = SH_base + ΔSH (only on object Gaussians; background untouched)
3. **Forward pass:** SH_modified ⟶ deformation network (warps to timestep t) ⟶ CUDA rasterizer (projects to camera i) ⟶ rendered image [3,H,W]
4. **Loss** (masked — only care about object pixels): L = L1(rendered × mask_2d, target × mask_2d)
5. **Backprop**: ∇L ⟶ rasterizer backward ⟶ deformation backward ⟶ ∇ΔSH (deformation weights are frozen — gradients pass through but only ΔSH gets updated)
6. **Gradient descent** on residual ∇ΔSH

Save source, target pairs to disk (~200 needed)



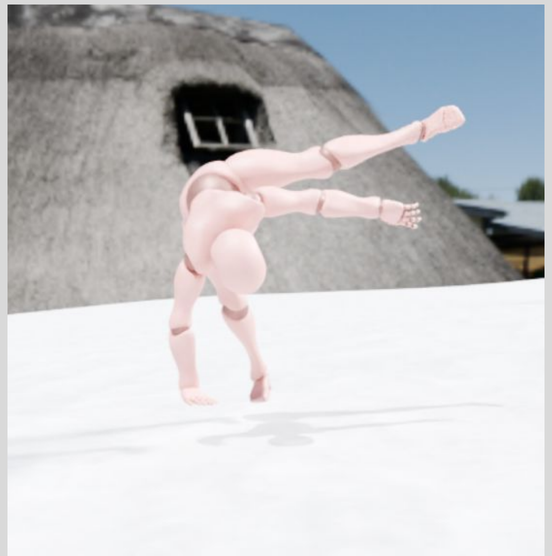
Bake & Export 3DGS/4DGS

Experiment 1: Blender Oracle

*Assuming a **perfect** Harmonizer that can match foreign object A's lighting to the background, can we optimize the spherical coefficients of Object A to match the background?*

Manually built Blender scene with **animated breakdancer** and **static cat**

- ➔ Breakdancer asset has *bright outdoor* HDRI (sunny, sky)
- ➔ Static cat asset has *dark forest* HDRI (moonlight, shaded)
- ➔ Treat snowy, white scene as background for juxtaposition



Breakdancer (sunny)

Cat (dark)

200 train camera views
40 test camera views

Unharmonized

3DGS & 4DGS

Generate source view for optimization

Compute loss & backpropagate through 3DGS/4DGS

Harmonized

3DGS & 4DGS

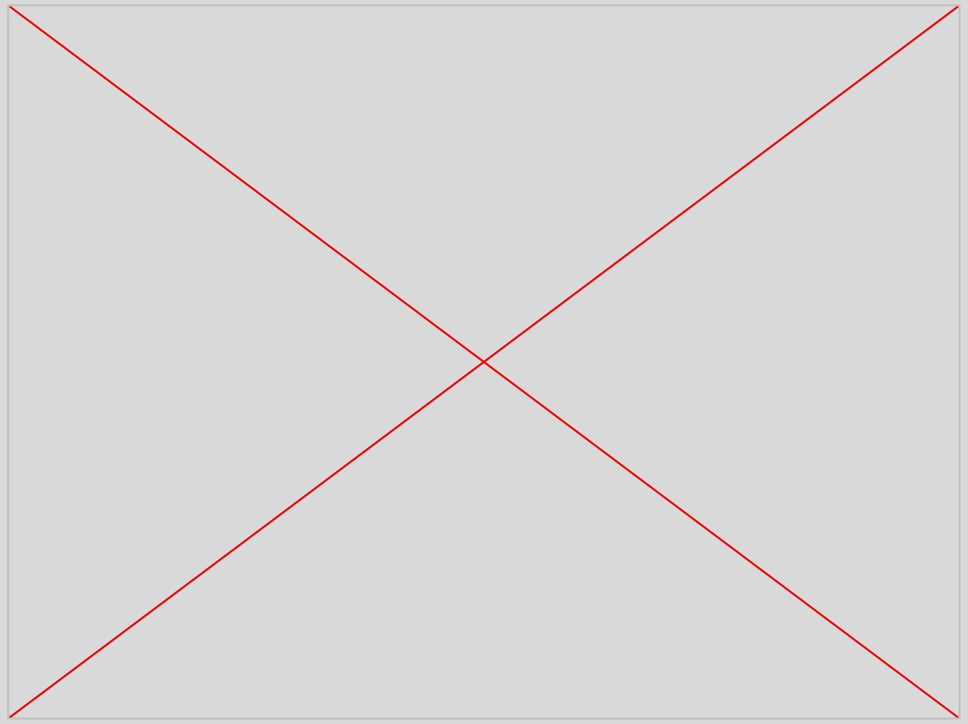
Generate target view for optimization

New Harmonizer Targets (Harmonized GS = ground truth harmonized views)

Experiment 1 Results

Unharmonized (3DGS)

TranSplat Baseline (3DGS)



psnr = 19.007

Harmonized 3D Gaussian (Our Method)

psnr = 20.109

Unharmonized 4DGS

Harmonized 4D Gaussian (Our Method)

Ground Truth 4DGS

Hypothesis: Modify only the DC band as in TranSplat + having separate LRs for different SH degrees

Harmonized 4D Gaussian

lpips = 0.2624

Separate LR Harmonization

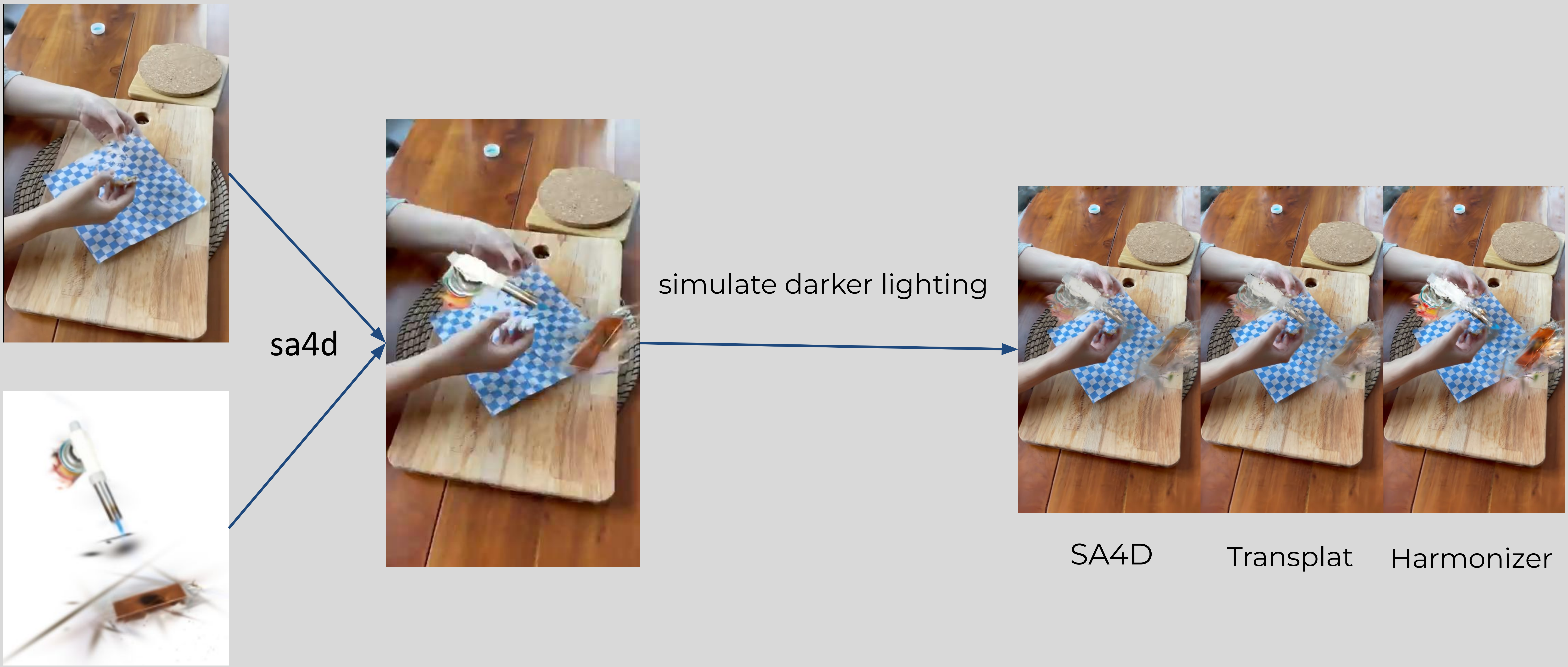
DC Band Optimization + Harmonization

lpips = 0.2214

Finding: optimizing only the DC band / specific SH's >> modifying all SH together at once

Experiment 2: Harmonization Pipeline on SA4D

How does our backpropagation pipeline work for composited real-life 4DGS scenes?



Results for Sa4D

Stretch Goals: Results for Multi-Object scenes and Difix3D+

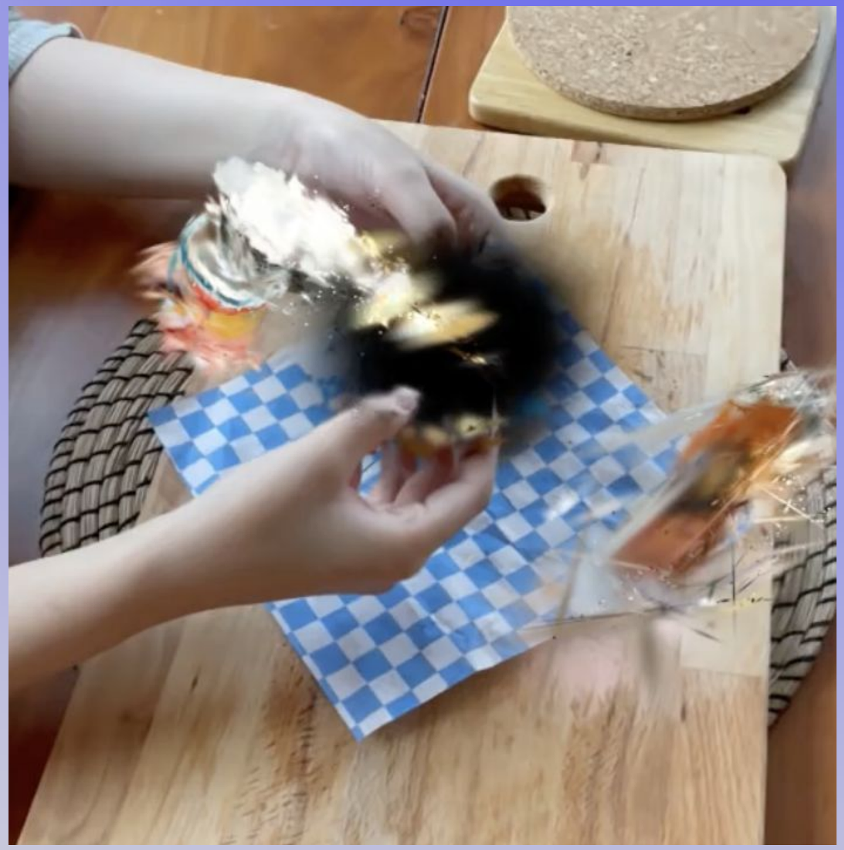
Improving the backpropagation pipeline for multi-object scenes and applying Difix3D+

Base 4DGS
multi-object scene

Harmonized
Multi-Object Scene

Harmonized Multi-Object
Scene w/Difix3D+

Future Goals



- **Improved masking quality**: Many of our renders suffer from masking being poor.
- **Specular Relighting:** Enhancing our 2D Harmonization to update Spherical Gaussians in order to allow the object to dynamically reflect light based on the environment
- **Dynamic Shadow Generation**: Testing different methods for shadow generation such as a shadow map or training an MLP to predict realistic shadows

07

Questions?